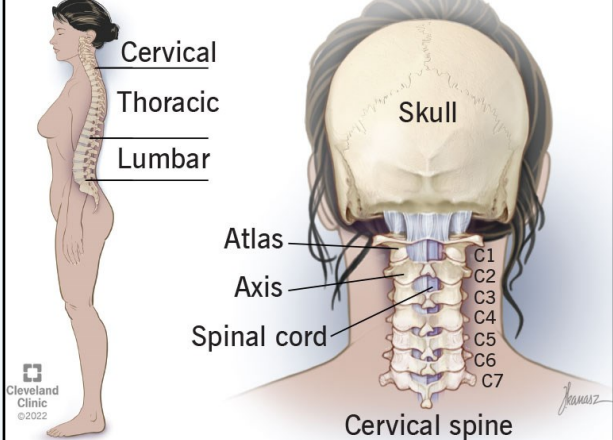




Evaluation of the Adult Patient with Neck Pain

Anatomy

Understanding the Cervical Spine



Note: From *Cervical Spine* [Infographic], by Cleveland Clinic, 2022, Cleveland Clinic (<https://my.clevelandclinic.org/health/articles/22278-cervical-spine>)

The cervical spine is comprised of seven vertebrae. The articulation between the occiput and the first cervical vertebra (the atlantooccipital joint) allows for approximately one-third of flexion and extension and one-half of lateral bending of the neck. The articulation between the first and second cervical vertebrae (the atlantoaxial joint) allows for 50 percent of rotational range of motion. The articulations between the second and seventh cervical vertebrae allow for approximately two-thirds of flexion and extension, 50 percent of rotation, and 50 percent of lateral bending.

The most common locations that exhibit degenerative changes are between C4 and C7. The nerve roots passing through the intervertebral foramina in these areas are C5, C6, and C7.

Differential Causes of Neck Pain

Causes of Atraumatic Neck Pain in Adults: Clinical Features and Supportive Findings

Condition	Symptoms	Physical Findings
Musculoskeletal		
Cervical strain	Pain and/or stiffness on neck movement	Tenderness or palpation of neck and trapezius muscles
"Whiplash" injury	Pain and/or stiffness on neck movement following an abrupt extension-flexion type injury (may present immediately or may be delayed for several days); other symptoms may include headache, shoulder or back pain, dizziness, paresthesia, fatigue, and sleep disturbances	Decreased range of motion associated with neck spasm
Cervical discogenic pain	Pain and/or stiffness on neck movement; cervical radicular symptoms are sometimes present (refer to cervical radiculopathy below)	Decreased range of motion with associated pain
Cervical facet osteoarthritis	Pain and/or stiffness on neck movement; symptoms can be somatically referred to the shoulders, periscapular region, occiput, or proximal limb	Decreased range of motion associated with neck spasm
Cervical myofascial pain	Focal pain and pressure sensitivity; often involves right side of neck and shoulder; pain typically has a deep, aching quality, occasionally accompanied by a sensation of burning or stinging	Localized tenderness ("trigger points") on palpation
Diffuse skeletal hyperostosis (DISH)	May also have thoracic spine, low back, and/or extremity pain; spinal morning stiffness is common; some affected patients may complain of dysphagia	Decreased range of motion may be present
Radiculopathy/Myelopathy		
Cervical radiculopathy	Pain, numbness, and/or tingling in a dermatomal distribution, and/or weakness in upper extremity	Decreased or altered sensation, diminished deep tendon reflexes, and/or decreased strength in upper extremity
Cervical spondylotic myelopathy	Lower extremity weakness, gait or coordination difficulties, and bowel or bladder dysfunction	Focal neurologic signs in upper and/or lower extremities may be present
Ossification of posterior longitudinal ligament	Typically present in the fifth to sixth decades of life with neck pain, stiffness, and progressive radiculopathy/myelopathy symptoms	Focal neurologic signs in upper and/or lower extremities may be present
Non-Spinal Causes		
Coronary artery disease (angina pectoris, MI)	Chest pain with radiation to neck; pain that worsens with exertion	Normal neck exam
Infection (osteomyelitis, discitis, abscess, pharyngeal meningitis)	Fever; other signs vary depending on nature of infection	Vary depending on nature of infection
Malignancy (metastatic)	Unexplained weight loss or prior history of cancer	Localized tenderness on palpation of spine
Neurologic Conditions		
Tension headache	Bilateral dull headache, which may be associated with neck pain; no other neurologic symptoms	Localized tenderness on palpation of scalp and/or neck; no neurologic abnormalities
Cervical dystonia	Sustained or intermittent muscle spasms of neck	Muscle contractions causing abnormal, often repetitive, movements and/or postures
Chiari malformation	Neck pain or headache from meningeal irritation is the most common presentation	Variable focal central nervous system signs
Referred shoulder pain (e.g., rotator cuff tear, adhesive capsulitis, impingement)	Shoulder pain with radiation to neck	Localized tenderness on shoulder exam with or without decreased range of motion

Musculoskeletal Conditions

Cervical strain: Cervical strain generally presents with pain and/or stiffness on neck movement. There is often a history of antecedent injury to the cervical paraspinal muscles, although it may result from the physical stresses of everyday, life including poor posture and sleeping habits.

Cervical spondylosis: "Spondylosis" is a nonspecific term used to describe effects generally ascribed to degenerative changes in the spine, usually with osteophyte production.

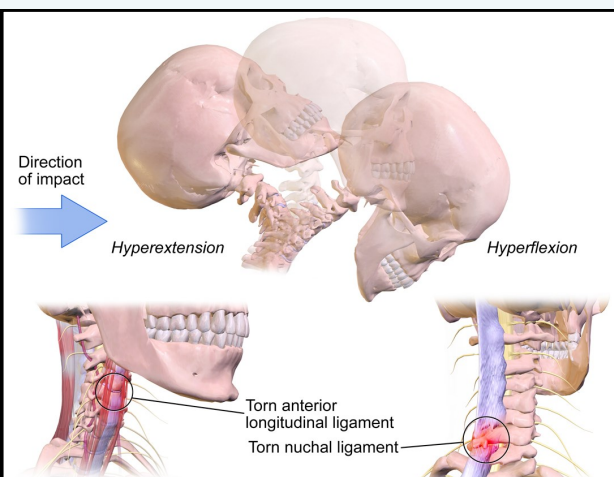
Cervical discogenic pain: Cervical discogenic pain results from disc degeneration. It typically presents with pain and/or stiffness on neck movement, which is sometimes associated with pain in the upper extremities. Symptoms are often exacerbated when the neck is held in one position for prolonged periods, such as occurs with driving, reading, or working at a computer.

Whiplash injury: Whiplash injury is defined as neck injury resulting from an acceleration-deceleration mechanism that causes sudden extension and flexion of the neck. These injuries are also commonly referred to as cervical strains or sprains.

Myofascial pain syndrome: Myofascial pain syndrome (MPS) is a regional pain disorder associated with trigger points, taut bands, and pressure sensitivity. MPS is a relatively common source of chronic pain in the general population.

Cervical radiculopathy: Cervical radiculopathy refers to dysfunction of the spinal nerve root. Cervical radiculopathy generally presents with pain, sensory abnormalities, and/or weakness in an upper extremity. Physical examination may show altered sensation in a dermatomal pattern, decreased reflexes, and/or localized muscle weakness in a myotomal distribution.

Cervical spondylotic myelopathy: Cervical spondylotic myelopathy refers to spinal cord injury or dysfunction caused by degenerative changes narrowing the spinal canal. Patients may present with a variety of neurologic complaints, including lower extremity weakness, gait or coordination difficulties, and bladder or bowel dysfunction.



Note: From *Whiplash* [Infographic], by BruceBlaus, 2015, Wikipedia ([https://en.wikipedia.org/wiki/Whiplash_\(medicine\)#/media/File:Whiplash.png](https://en.wikipedia.org/wiki/Whiplash_(medicine)#/media/File:Whiplash.png))

Evaluation

It is not always possible to reliably determine the specific cause of neck pain, nor is it necessary to do so in many cases. The initial evaluation focuses largely on excluding serious conditions that may require intervention.

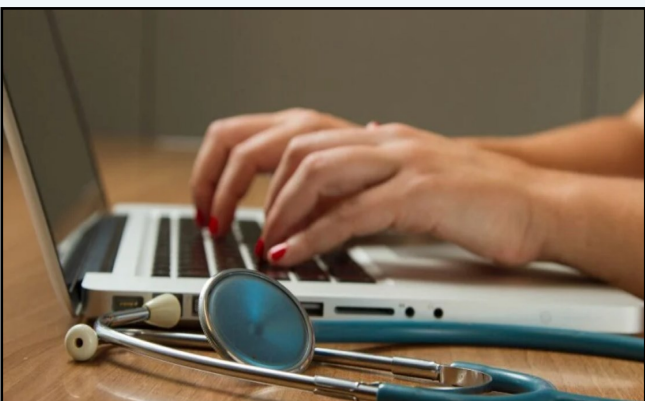
Most patients with atraumatic neck pain without red flags do **not** require imaging.

Only a minority of patients with nontraumatic neck pain require imaging. Imaging is generally indicated in the following circumstances:

- Serious, persistent, or progressive neurologic findings suggesting spinal cord compression, myelopathy, or severe radiculopathy
- Constitutional symptoms (fevers, chills, unexplained weight loss)
- Infectious risk (e.g., injection drug use, immunosuppression)
- History of malignancy
- Persistent moderate to severe neck pain (e.g., lasting >6 weeks and affecting sleep or ability to perform daily activities and/or occupation)

Physical examination includes observation of neck movement, palpation of the trapezius and paraspinal muscles, neurologic assessment for radicular and upper motor neuron signs, and provocative maneuvers in patients with radicular symptoms.

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References

BruceBlaus (2015). *Whiplash* [Infographic]. Wikipedia. [https://en.wikipedia.org/wiki/Whiplash_\(medicine\)#/media/File:Whiplash.png](https://en.wikipedia.org/wiki/Whiplash_(medicine)#/media/File:Whiplash.png)

Cleveland Clinic (2022). *Cervical Spine* [Infographic]. Cleveland Clinic. <https://my.clevelandclinic.org/health/articles/22278-cervical-spine>